

练习十二 波动（二）

1. $2\rho u\pi^2 v^2 A^2$, $120\rho u\pi^2 v^2 A^2 S$

2. 频率, 强度, 100, 13

3. (2); 4. (1)

5. (1) $\lambda = uT = 0.2 \times 1 = 0.2 \text{ m}$

$$\Delta\varphi = \varphi_2 - \varphi_1 - \frac{2\pi}{\lambda}(r_2 - r_1) = \pi - \frac{2\pi}{0.2}(0.5 - 0.4) = 0;$$

(2) $A = A_1 + A_2 = 0.2 + 0.2 = 0.4 \text{ (m)}$

6. 设波函数为 $y = A\cos(\omega t - \frac{2\pi}{\lambda}x + \varphi)$

即 $y = 0.1\cos(2\pi t - \frac{2\pi}{\lambda}x + \varphi)$

由质点 a 的初始条件得:

$$\begin{aligned} 0.1\cos(2\pi - \frac{2\pi}{\lambda} \times 0.1 + \varphi) &= 0 \\ -0.1 \times 2\pi \sin(2\pi - \frac{2\pi}{\lambda} \times 0.1 + \varphi) &< 0 \end{aligned}$$

由上面二式求得: $2\pi - \frac{2\pi}{\lambda} \times 0.1 + \varphi = 2k_a\pi + \frac{\pi}{2}$

由质点 b 的初始条件得:

$$\begin{aligned} 0.1\cos(2\pi - \frac{2\pi}{\lambda} \times 0.2 + \varphi) &= 0.05 \\ -0.1 \times 2\pi \sin(2\pi - \frac{2\pi}{\lambda} \times 0.2 + \varphi) &> 0 \end{aligned}$$

由上面二式求得: $2\pi - \frac{2\pi}{\lambda} \times 0.1 + \varphi = 2k_b\pi - \frac{\pi}{3}$

解得: $\lambda = 0.24 \text{ m},$

$$\varphi = \frac{4\pi}{3}$$

波函数为 $y = 0.1\cos(2\pi t - \frac{2\pi}{0.24}x + \frac{4\pi}{3})$